DON BOSCO INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

under DBIT_ SIC

INDUSTRY SAFETY MONITORING SYSTEM

SAMSUNG Innovation Campus Smart Things Edition

PROBLEM STATEMENT

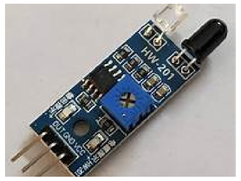
Conveyor belt operations are prone to hazards like fire, gas leaks, and overheating, leading to safety risks and disruptions. There is a need for a real-time monitoring system to detect hazards, trigger alerts, and take preventive actions.

OBJECTIVE

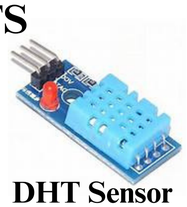
- Detect hazards in real time, triggering an LED alert and stopping the conveyor.
- Enable cloud-based monitoring for transparency and safer operations.

KEY COMPONENTS

Esp 32



IR Sensor



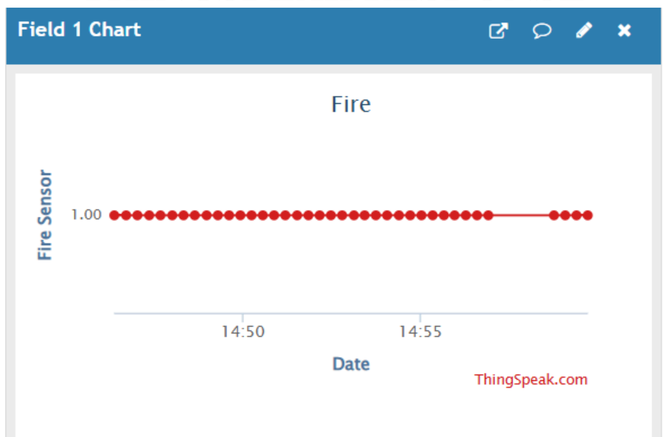
DHT Sensor

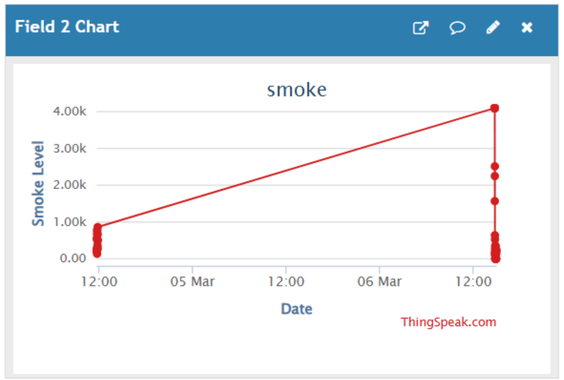
Raspberry Pi

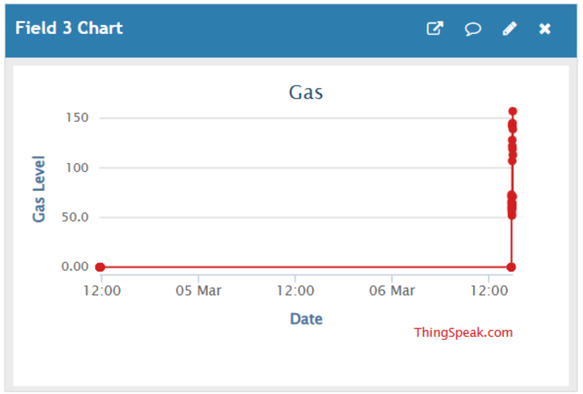
Fire Sensor

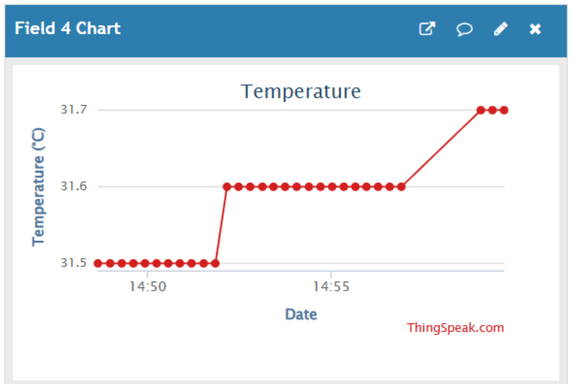
Mq5 Sensor

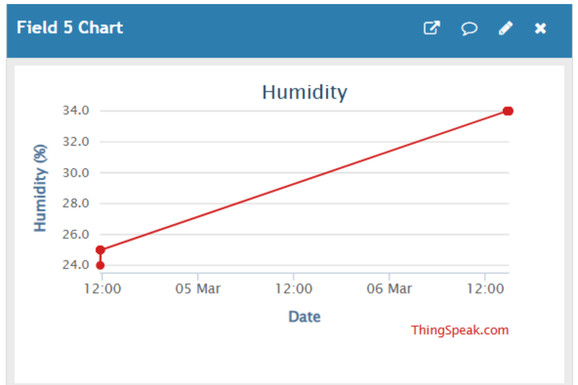
THINGSPEAK RESPONSE

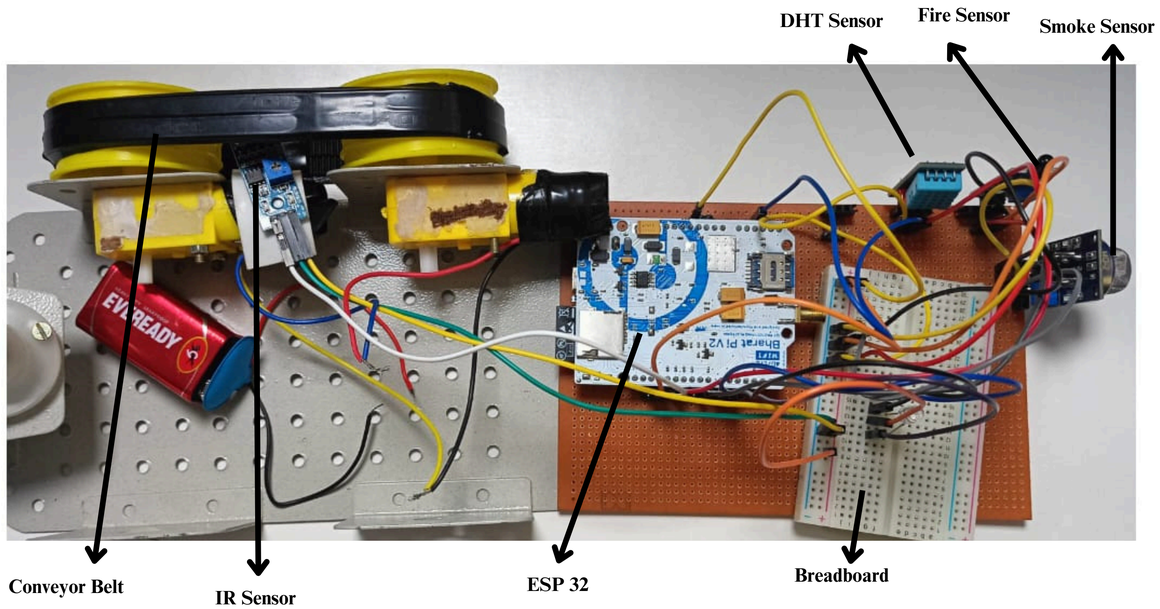




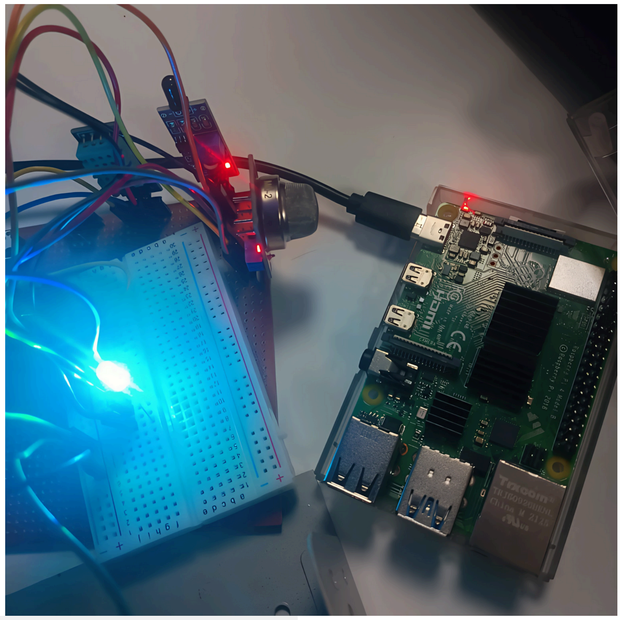


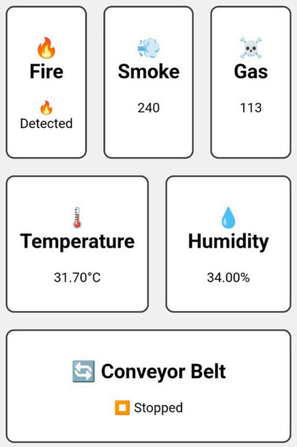


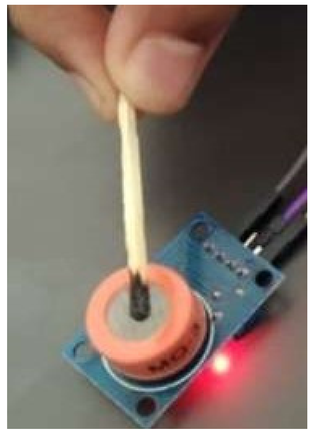




WORKING







CONCLUSION

The system uses ESP32 to sense sensors triggering alerts and stopping the conveyor. Data is sent to ThingSpeak for monitoring, while Raspberry Pi acts as a server to display values.

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